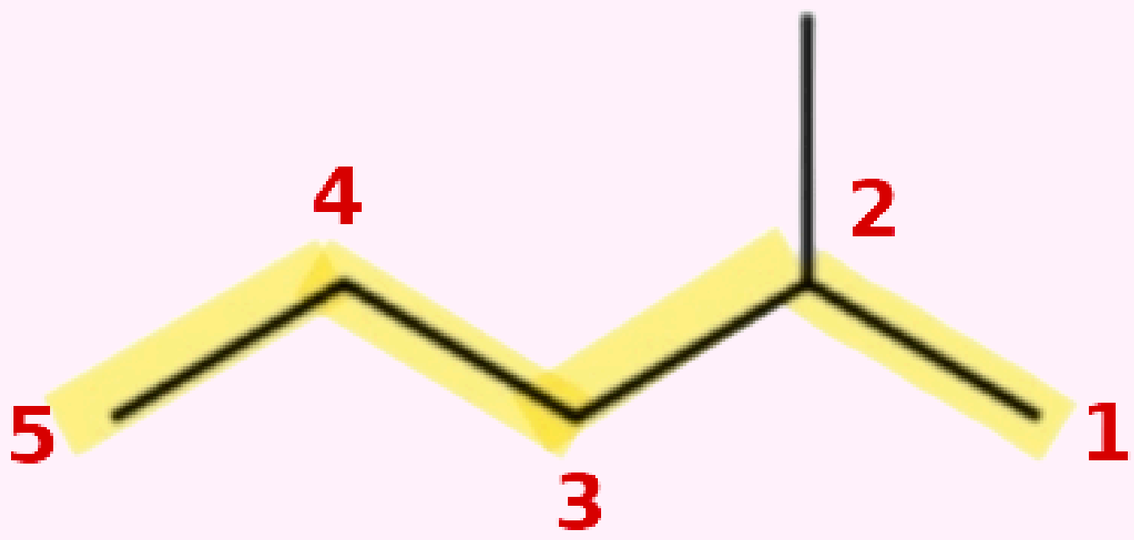
# CARD A01 — STRUCTURE → NAME

**Level: FOUNDATION** **Molecular formula: $C_6H_{14}$**

## 1. NUMBERED PARENT




## 2. NAMING TRAIL

**Parent:** longest chain = 5 C → pentane. Several 5-C paths exist (the top methyl can serve as a chain end), but each carries one methyl group, so they lead to the same name.

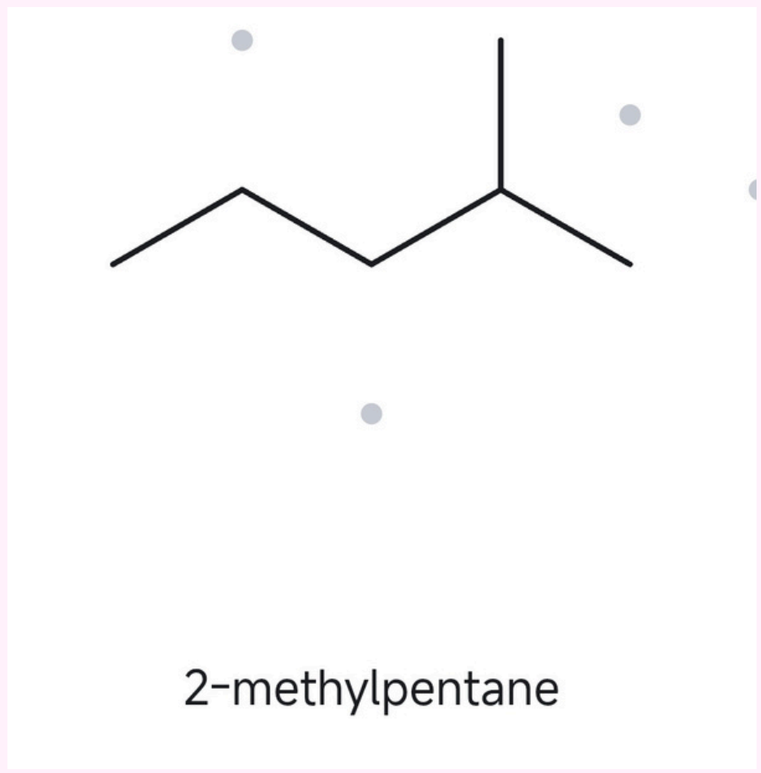
**Numbering:** left → right gives 4; right → left gives 2. First point of difference: 2 < 4, so number from the right

**Substituents:** methyl on C2 → 2-methylpentane.

## 3. MY NAME

**2-methylpentane**

## 4. THE ALKANE CHECK



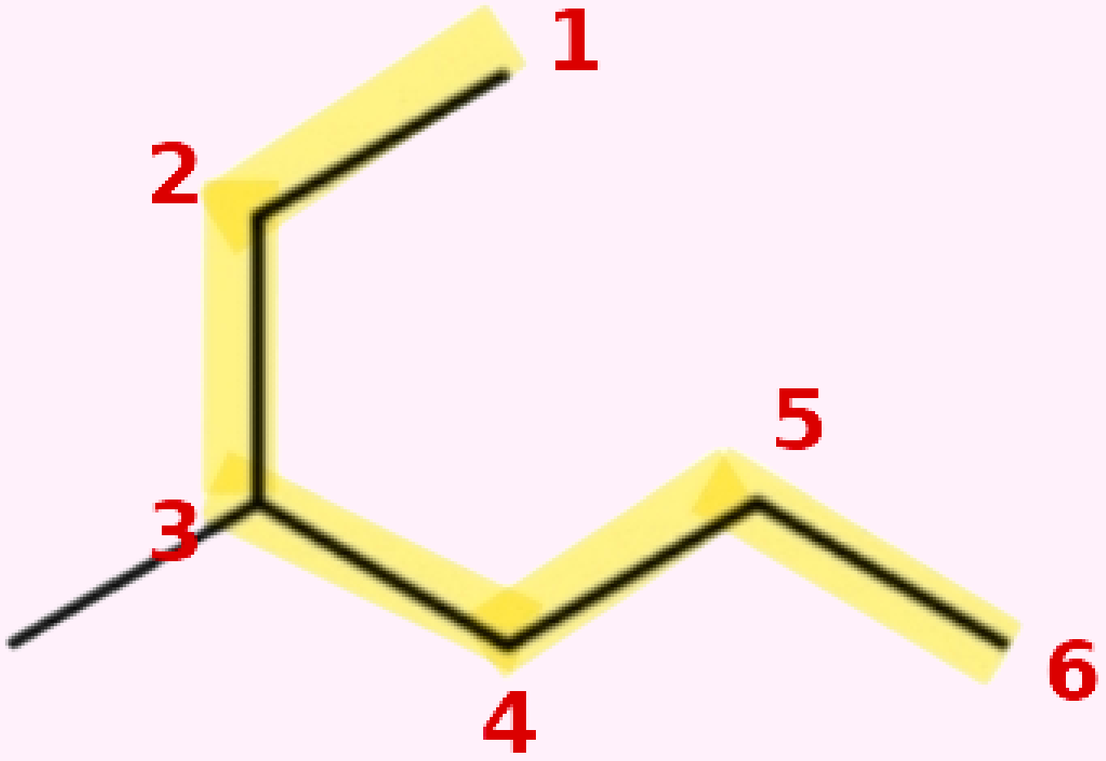

**MATCH**

- [x] longest chain confirmed
- [x] lowest locants
- [x] alphabetical order
- [x] punctuation
- [x] formula counted

# CARD A02 — STRUCTURE → NAME

**Level: FOUNDATION** **Molecular formula: $C_7H_{16}$**

## 1. NUMBERED PARENT




## 2. NAMING TRAIL

**Parent:** longest chain = 6 C → hexane. Only one 6-C path exists. Any path through the methyl branch reaches at most 5 C, so there is no competing chain.

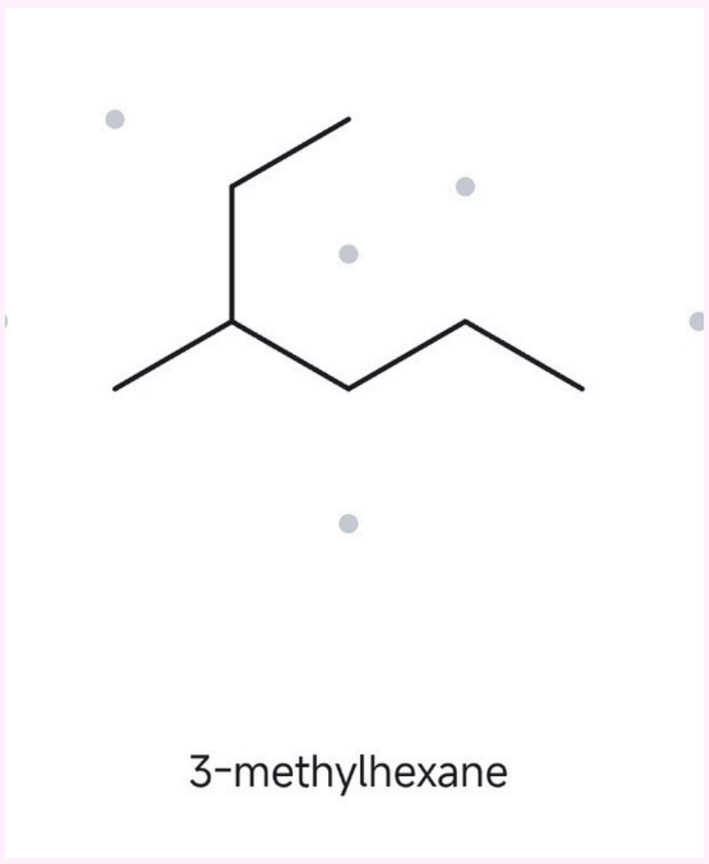
**Numbering:** top → bottom right gives 3; bottom right → top gives 4. First point of difference: 3 < 4, so number from the top.

**Substituents:** methyl on C3 → 3-methylhexane.

## 3. MY NAME

# 3-methylhexane

## 4. THE ALKANE CHECK



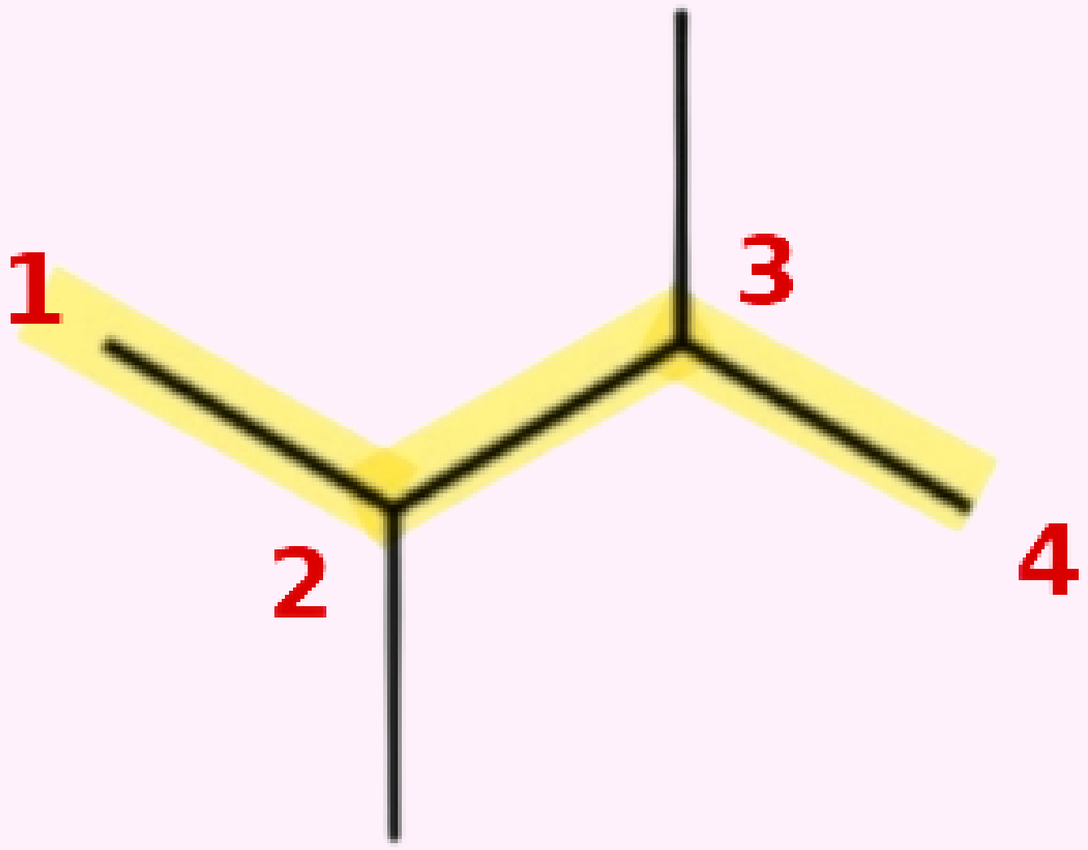

**MATCH**

- [x] longest chain confirmed
- [x] lowest locants
- [x] alphabetical order
- [x] punctuation
- [x] formula counted

# CARD A03 — STRUCTURE → NAME

**Level: FOUNDATION** **Molecular formula: $C_6H_{14}$**

## 1. NUMBERED PARENT




## 2. NAMING TRAIL

**Parent:** longest chain = 4 C → butane. Several 4-C paths exist (either methyl can serve as a chain end), but each carries two methyl groups, so they lead to the same name.

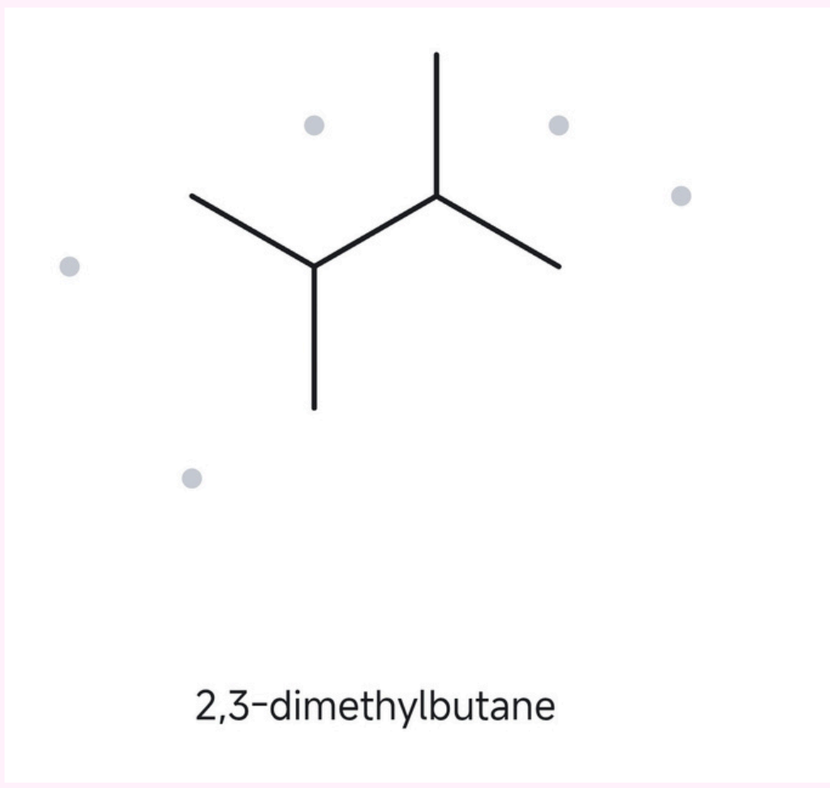
**Numbering:** left → right gives 2,3; right → left gives 2,3. The two directions tie at every point, so there is no first point of difference and either end works. I numbered from the left.

**Substituents:** methyl on C2 and C3 → 2,3-dimethyl.

## 3. MY NAME

**2,3-dimethylbutane**

## 4. THE ALKANE CHECK



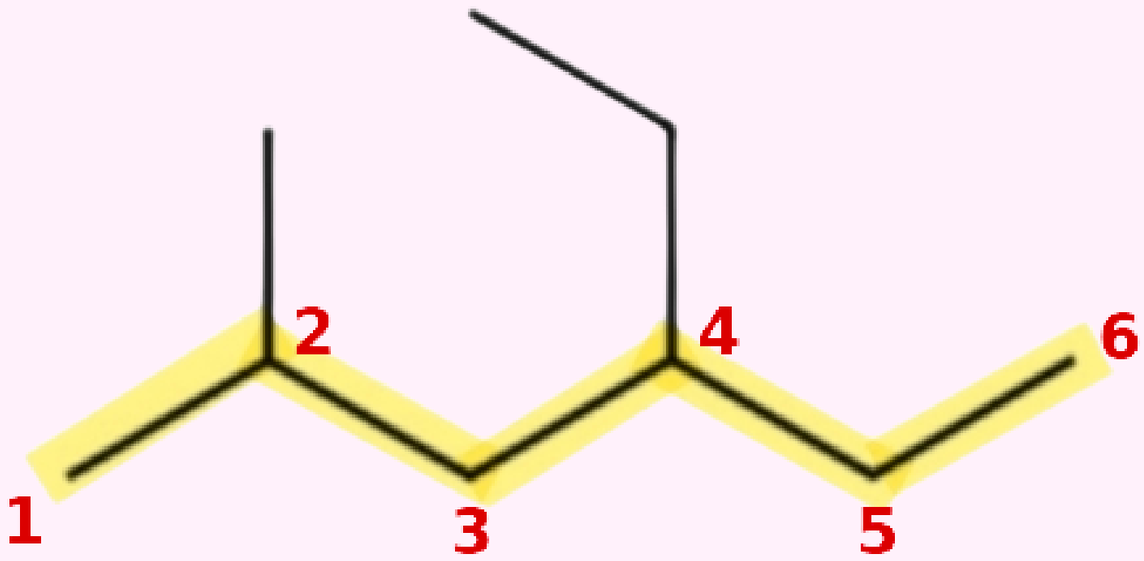

**MATCH**

- [x] longest chain confirmed
- [x] lowest locants
- [x] alphabetical order
- [x] punctuation
- [x] formula counted

# CARD A04 — STRUCTURE → NAME

**Level: DEVELOPING** **Molecular formula: $C_9H_{20}$**

## 1. NUMBERED PARENT




## 2. NAMING TRAIL

**Parent:** longest chain = 6 C → hexane. Several 6-C paths exist (the methyl or the ethyl's end can serve as a chain end), but each carries one methyl and one ethyl group, so they lead to the same name.

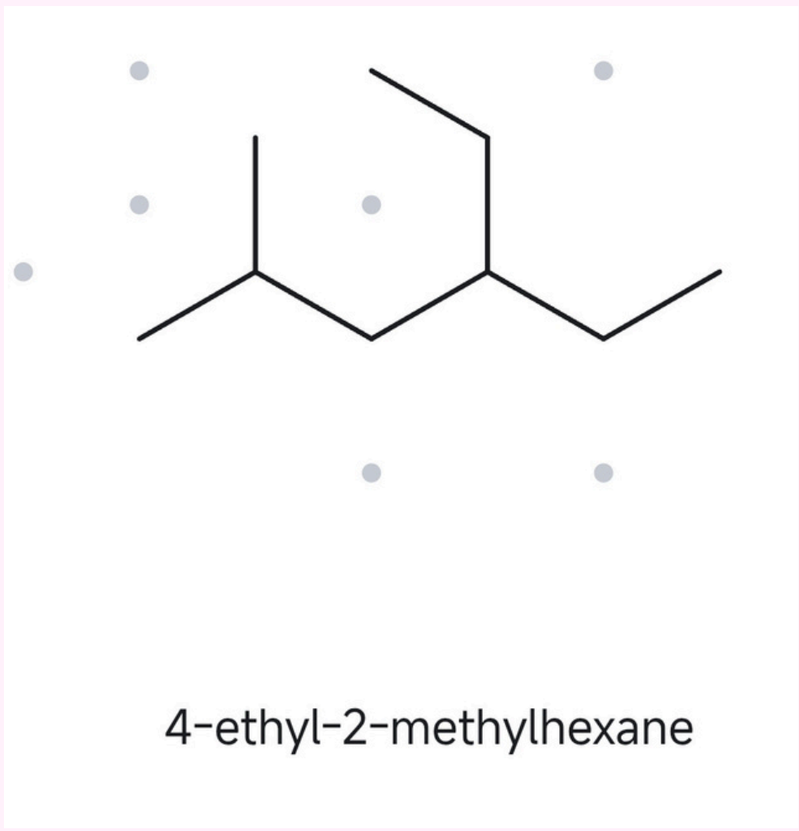
**Numbering:** left → right gives 2,4; right → left gives 3,5. First point of difference: 2 < 3, so number from the left.

**Substituents:** methyl on C2 and ethyl on C4. Listed alphabetically (ethyl before methyl) → 4-ethyl-2-methyl.

## 3. MY NAME

**4-ethyl-2-methylhexane**

## 4. THE ALKANE CHECK



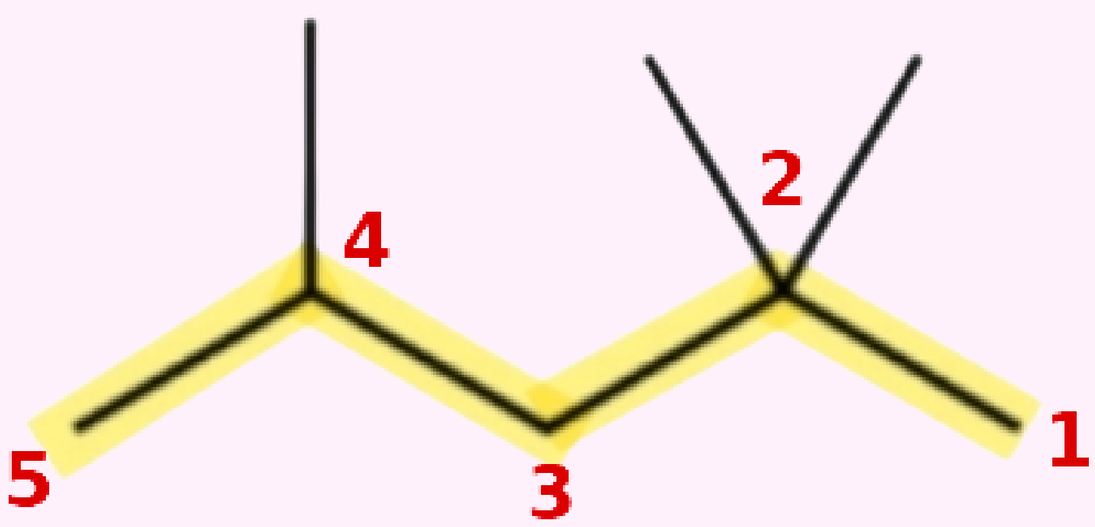

MATCH

- [x] longest chain confirmed
- [x] lowest locants
- [x] alphabetical order
- [x] punctuation
- [x] formula counted

# CARD A05 — STRUCTURE → NAME

**Level: DEVELOPING** **Molecular formula: $C_8H_{18}$**

## 1. NUMBERED PARENT




## 2. NAMING TRAIL

**Parent:** longest chain = 5 C → pentane. Several 5-C paths exist (any methyl on C2 or C4 can serve as a chain end), but each carries three methyl groups, so they lead to the same name.

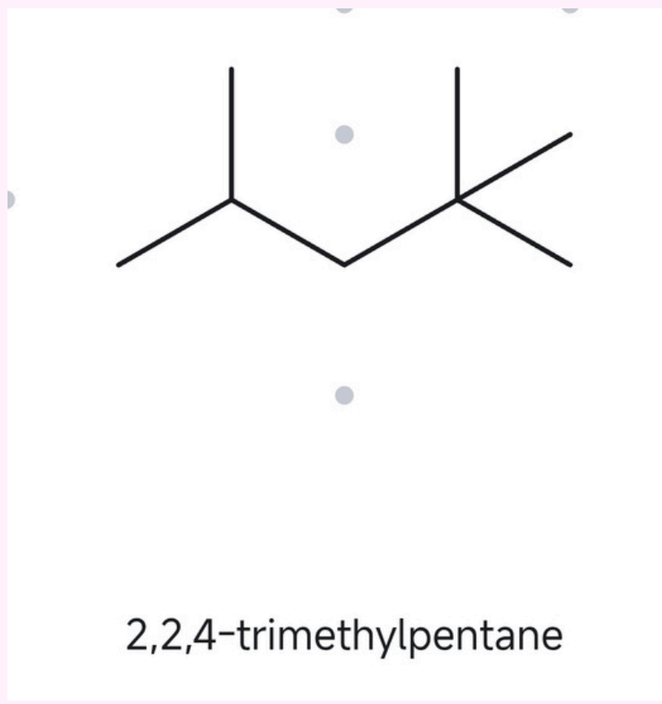
**Numbering:** left → right gives 2,4,4; right → left gives 2,2,4. First point of difference: the first locants tie at 2, but at the second position 2 < 4, so number from the right.

**Substituents:** methyl on C2, C2 and C4 → 2,2,4-trimethyl.

## 3. MY NAME

**2,2,4-trimethylpentane**

## 4. THE ALKANE CHECK



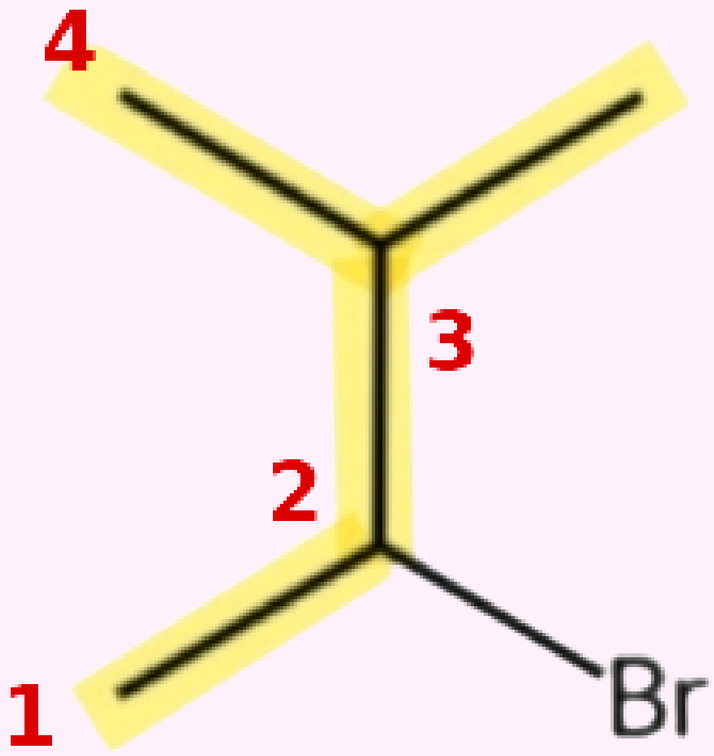

MATCH

- [x] longest chain confirmed
- [x] lowest locants
- [x] alphabetical order
- [x] punctuation
- [x] formula counted

# CARD A06 — STRUCTURE → NAME

**Level: DEVELOPING** **Molecular formula: $C_5H_{11}Br$**

## 1. NUMBERED PARENT




## 2. NAMING TRAIL

**Parent:** longest chain = 4 C → butane. Two 4-C paths exist (either top arm can serve as the chain end), but each carries one methyl and one bromine, so they lead to the same name.

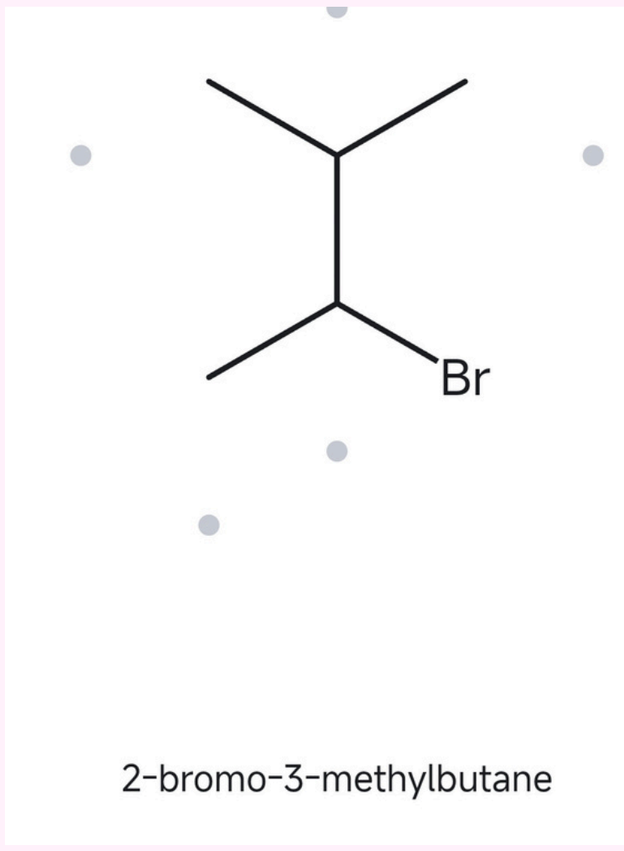
**Numbering:** bottom → top gives 2,3 (Br on C2, methyl on C3); top → bottom gives 2,3 (methyl on C2, Br on C3). The locant sets tie at every point, so the first point of difference does not decide. Tie-breaker: alphabetical order. Bromo comes before methyl, so Br gets the lower number, and we number from the bottom.

**Substituents:** bromo on C2 and methyl on C3. Listed alphabetically (bromo before methyl) → 2-bromo-3-methyl.

## 3. MY NAME

**2-bromo-3-methylbutane**

## 4. THE ALKANE CHECK



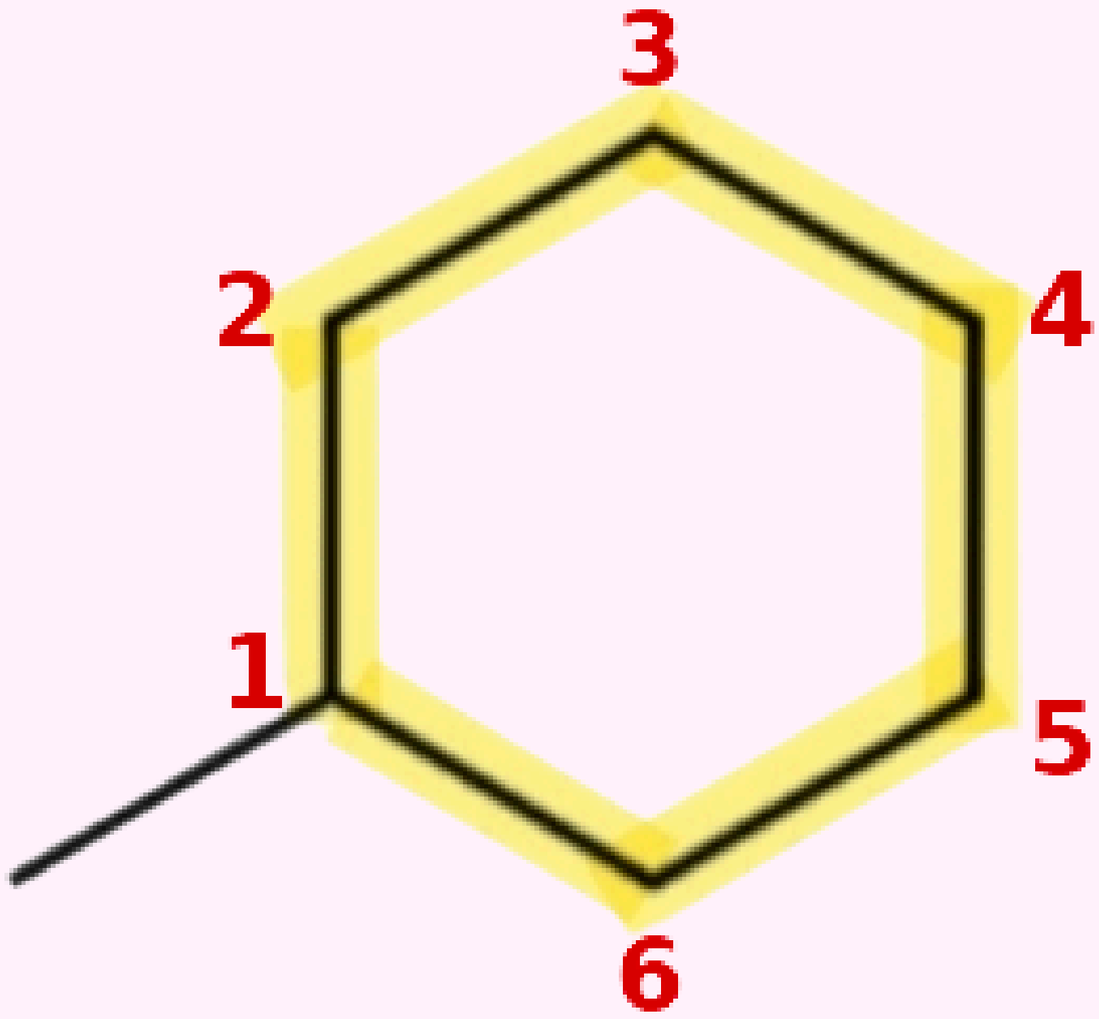

MATCH

- [x] longest chain confirmed
- [x] lowest locants
- [x] alphabetical order
- [x] punctuation
- [x] formula counted

# CARD A07 — STRUCTURE → NAME

**Level: DEVELOPING** **Molecular formula: $C_7H_{14}$**

## 1. NUMBERED PARENT




## 2. NAMING TRAIL

**Parent:** the ring has 6 C → cyclohexane. The ring (6 C) is larger than the substituent (1 C), so the ring is the parent, and there is only one possible ring path.

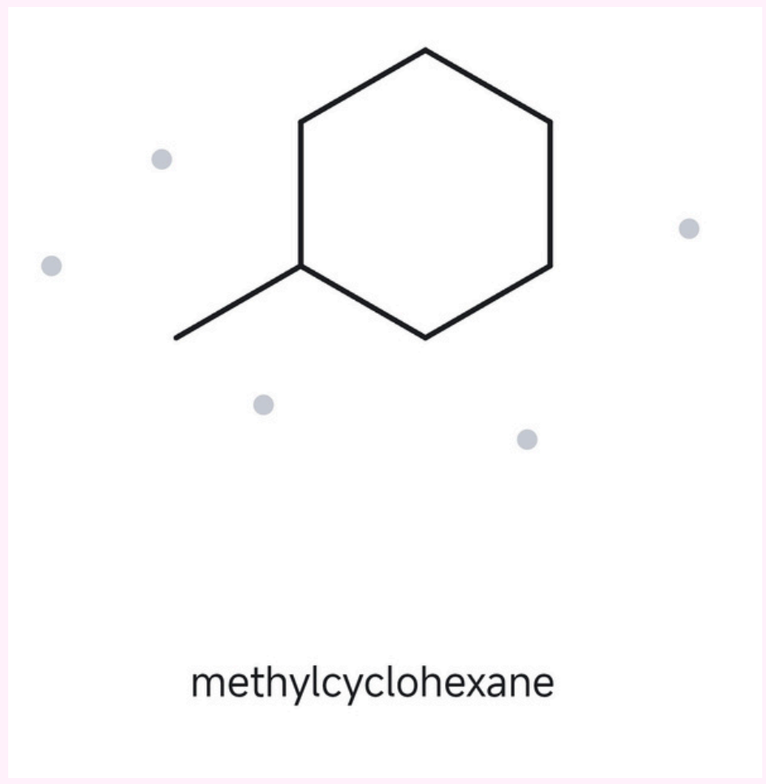
**Numbering:** the ring carbon bearing the methyl is C1. Clockwise or anticlockwise, the single substituent gets locant 1 either way. The two directions tie, so there is no first point of difference, and either direction works. A lone substituent on a ring needs no locant in the name.

**Substituents:** methyl on C1 → methyl.

## 3. MY NAME

**Methylcyclohexane**

## 4. THE ALKANE CHECK



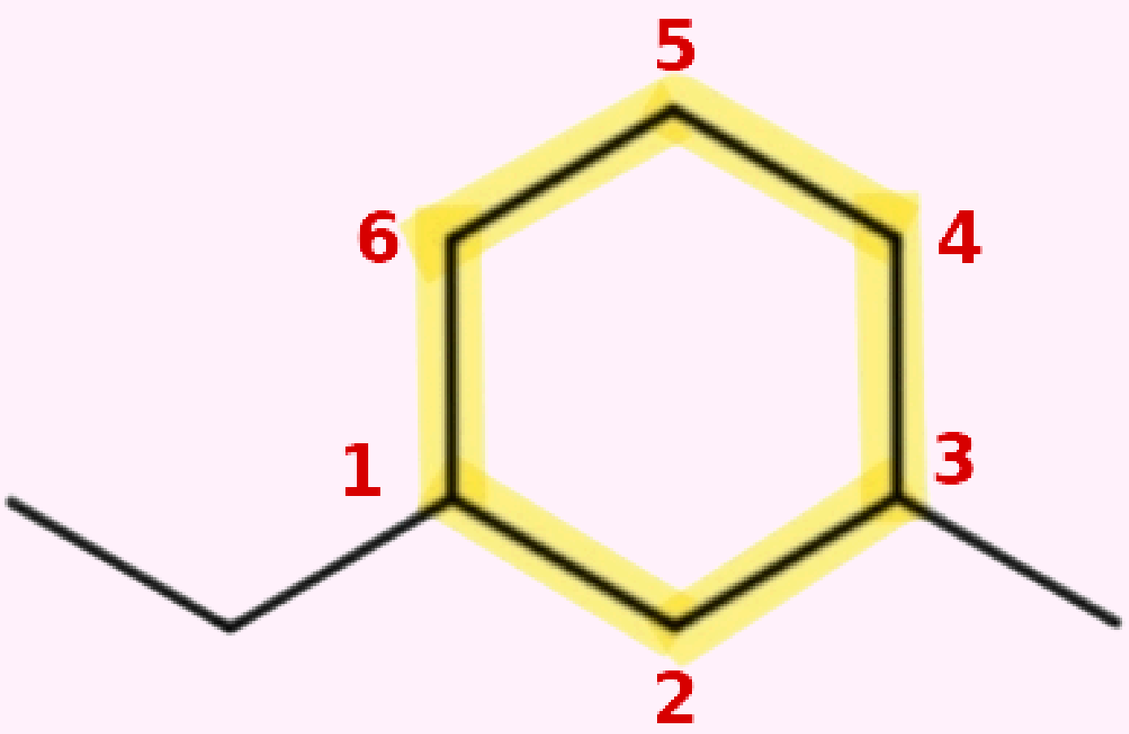

MATCH

- [x] longest chain confirmed
- [x] lowest locants
- [x] alphabetical order
- [x] punctuation
- [x] formula counted

# CARD A08 — STRUCTURE → NAME

**Level: CHALLENGE** **Molecular formula: $C_9H_{18}$**

## 1. NUMBERED PARENT




## 2. NAMING TRAIL

**Parent:** the ring has 6 C → cyclohexane. The ring (6 C) is larger than either substituent (2 C and 1 C), so the ring is the parent, and there is only one possible ring path.

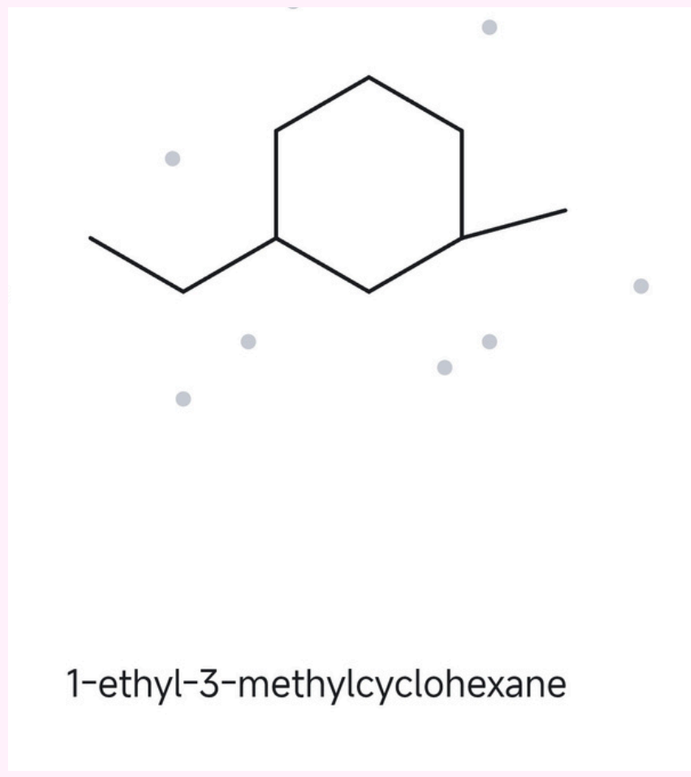
**Numbering:** the two substituents are separated by one ring carbon. Going through the bottom carbon gives 1,3; going the long way around gives 1,5. First point of difference: 3 < 5, so number through the bottom. Starting at the ethyl or at the methyl gives the same set (1,3), so that choice is a tie. Tie-breaker: alphabetical order. Ethyl comes before methyl, so the ethyl carbon is C1.

**Substituents:** ethyl on C1 and methyl on C3. Listed alphabetically (ethyl before methyl) → 1-ethyl-3-methyl.

## 3. MY NAME

**1-ethyl-3-methylcyclohexane**

## 4. THE ALKANE CHECK



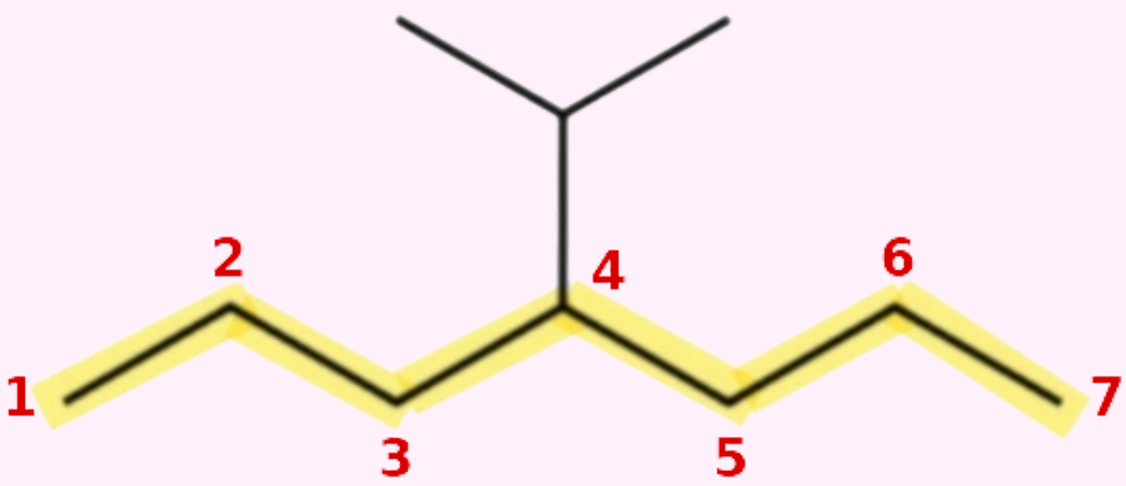

MATCH

- [x] longest carbon chain
- [x] lowest locants
- [x] alphabetical order
- [x] punctuation
- [x] formula counted

# CARD A09 — STRUCTURE → NAME

**Level: CHALLENGE** **Molecular formula: $C_{10}H_{22}$**

## 1. NUMBERED PARENT




## 2. NAMING TRAIL

**Parent:** longest chain = 7 C → heptane. Only one 7-C path exists. Any path through the isopropyl branch reaches at most 6 C, so there is no competing chain.

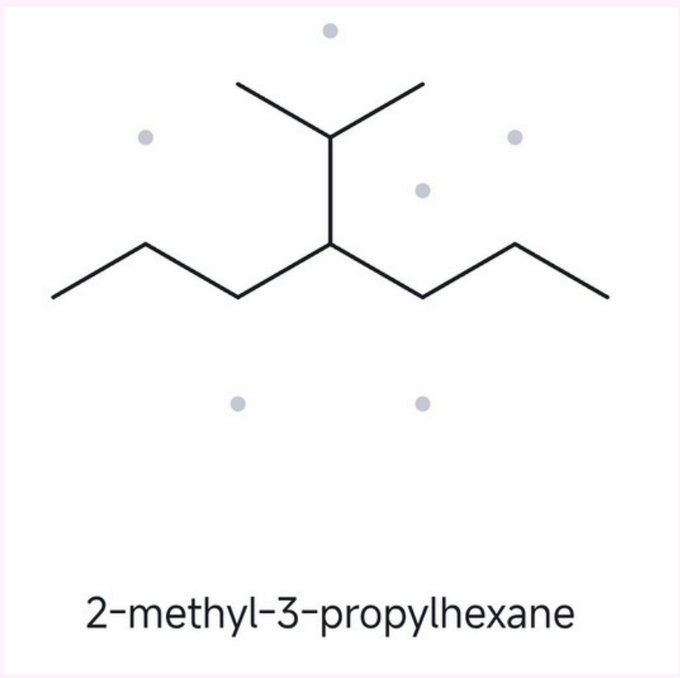
**Numbering:** left → right gives 4; right → left gives 4. The directions tie at every point, so there is no first point of difference and either end works. I numbered from the left.

**Substituents:** isopropyl (a 3-C branch, CH with two $CH_3$) on C4 → 4-isopropyl.

## 3. MY NAME

**4-isopropylheptane**

## 4. THE ALKANE CHECK




**DID NOT MATCH**

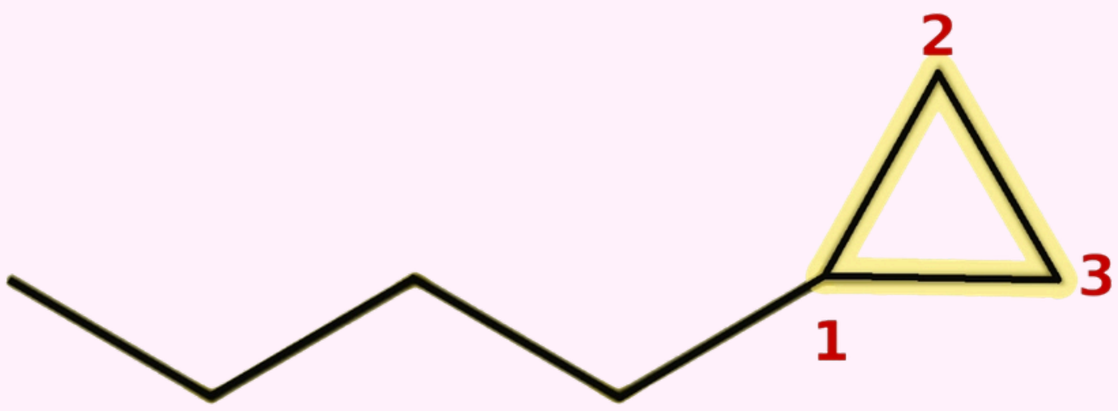
**Same compound ($C_{10}H_{22}$), but the tool chose a 6-C parent (hexane) when a 7-C chain exists. The longest chain must be the parent, so 4-isopropylheptane is correct.**

- [x] longest chain confirmed
- [x] lowest locants
- [x] alphabetical order
- [x] punctuation
- [x] formula counted

# CARD A10 — STRUCTURE → NAME

**Level: CHALLENGE** **Molecular formula: $C_7H_{14}$**

## 1. NUMBERED PARENT




## 2. NAMING TRAIL

**Parent:** the ring (3 C) → cyclopropane. Under the current IUPAC rule, a ring is the parent over a chain, regardless of size. The 4-C group is the substituent.

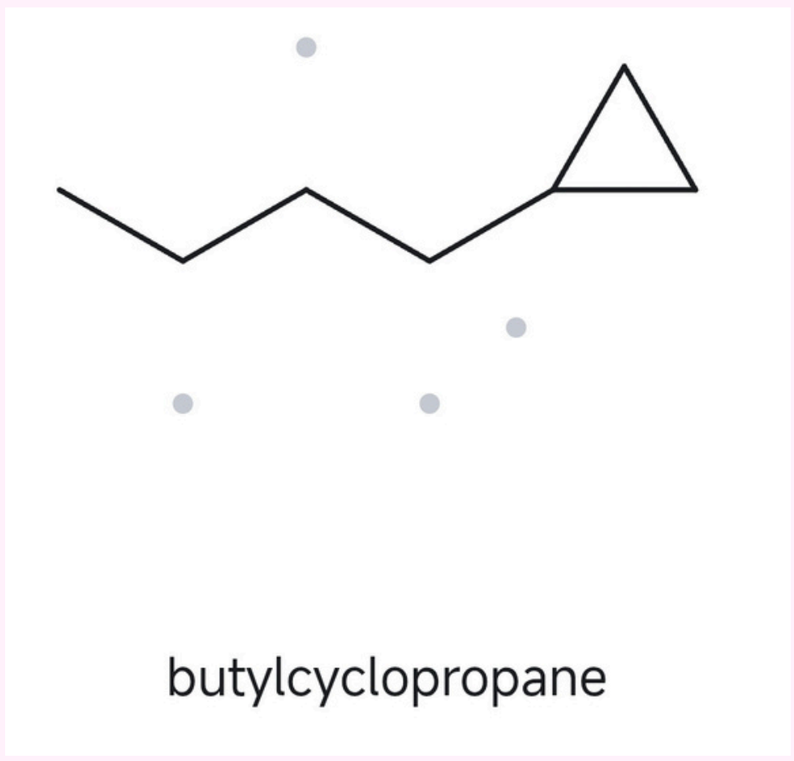
**Numbering:** the ring carbon bearing the substituent is C1. A single substituent gets locant 1 in either direction around the ring, so the directions tie, and no locant is written for it.

**Substituents:** a 4-carbon chain, attached through its end carbon → butyl.k

## 3. MY NAME

**butylcyclopropane**

## 4. THE ALKANE CHECK




**MATCH**

- [x] longest chain confirmed
- [x] lowest locants
- [x] alphabetical order
- [x] punctuation
- [x] formula counted

# CARD B01 — NAME → STRUCTURE

**Level: FOUNDATION** **GIVEN NAME: 2-methylpropane**

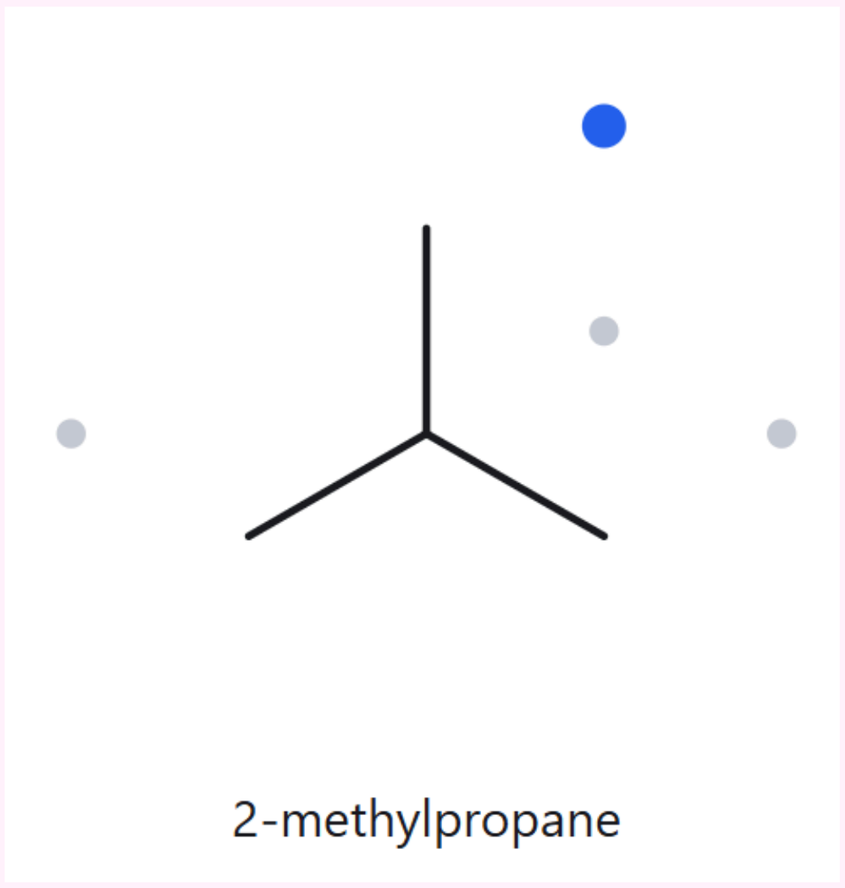
## 1. NAME DECODED

**Parent:**
propane → a 3-carbon chain

**Substituents:**
methyl on C2

**Expected formula:**
3 + 1 = 4 C; alkane → $C_4H_{10}$

## 2. BOND-LINE STRUCTURE (THE ALKANE)



2-methylpropane

## 3. FORMULA CHECK

**Counted from the drawing:**
$C_4H_{10}$. The drawing has 4 carbons (3 chain + 1 methyl), with 3 + 1 + 3 + 3 = 10 H

**Alkane:**
$C_nH_{2n+2} = C_4H_{10}$ ✓

## 4. BACK-CHECK

**Returned by The Alkane:**
2-methylpropane

✓ Same as the given name

- [x] parent length matches
- [x] substituent positions match
- [x] formula agrees
- [x] back-check recorded

# CARD B02 — NAME → STRUCTURE

**Level: FOUNDATION** **GIVEN NAME: 3-ethylpentane**

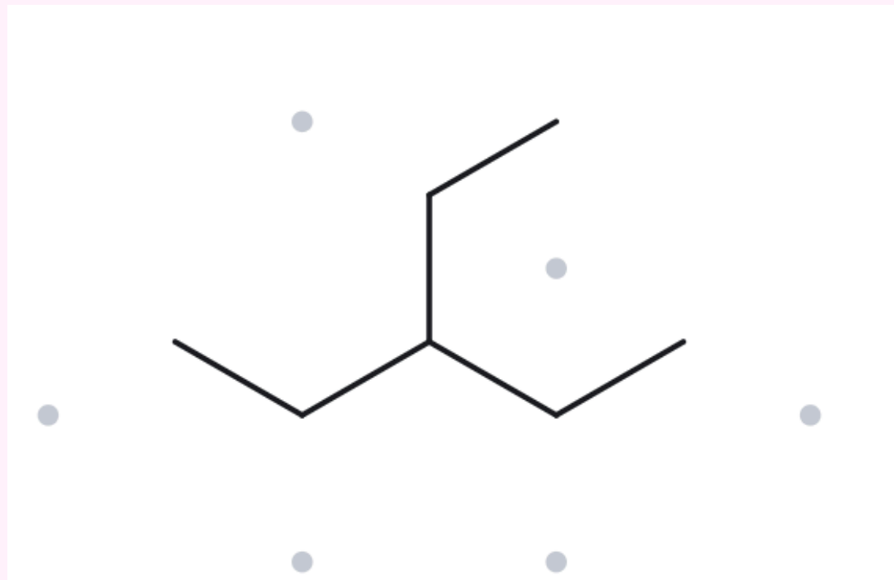
## 1. NAME DECODED

**Parent:**
pentane → a 5-carbon chain

**Substituents:**
ethyl on C3

**Expected formula:**
5 + 2 = 7 C; alkane → $C_7H_{16}$

## 2. BOND-LINE STRUCTURE (THE ALKANE)



3-ethylpentane

## 3. FORMULA CHECK

**Counted from the drawing:**
$C_7H_{16}$ The drawing has 7 carbons (5 chain + 2 ethyl). The H count is 3 + 2 + 1 + 2 + 3 = 11 on the chain, plus 5 on the ethyl, which gives 16 H

**Alkane:**
$C_nH_{2n+2} = C_7H_{16}$ ✓

## 4. BACK-CHECK

**Returned by The Alkane:**
3-ethylpentane

✓ Same as the given name

- [x] parent length matches
- [x] substituent positions match
- [x] formula agrees
- [x] back-check recorded

# CARD B03 — NAME → STRUCTURE

**Level: FOUNDATION** **GIVEN NAME: 2,2-dimethylbutane**

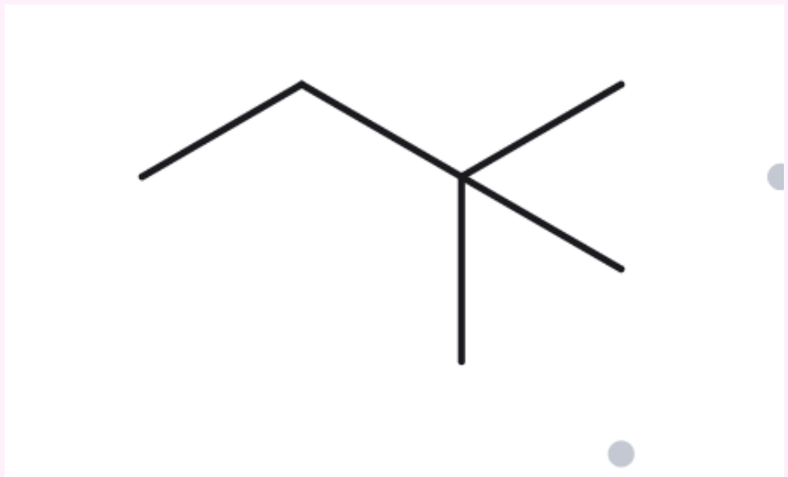
## 1. NAME DECODED

**Parent:**
butane → a 4-carbon chain

**Substituents:**
two methyl groups, both on C2

**Expected formula:**
4 + 2 = 6 C; alkane → $C_6H_{14}$

## 2. BOND-LINE STRUCTURE (THE ALKANE)



2,2-dimethylbutane

## 3. FORMULA CHECK

**Counted from the drawing:**
$C_6H_{14}$. The drawing has 6 carbons (4 chain + 2 methyl). The H count on the chain is 3 + 0 + 2 + 3 = 8 (C2 has no H), plus 3 + 3 = 6 on the two methyls, which gives 14 H

**Alkane:**
$C_nH_{2n+2} = C_6H_{14}$ ✓

## 4. BACK-CHECK

**Returned by The Alkane:**
2,2-dimethylbutane

✓ Same as the given name

- [x] parent length matches
- [x] substituent positions match
- [x] formula agrees
- [x] back-check recorded

# CARD B04 — NAME → STRUCTURE

**Level: DEVELOPING** **GIVEN NAME: 3-ethyl-2-methylpentane**

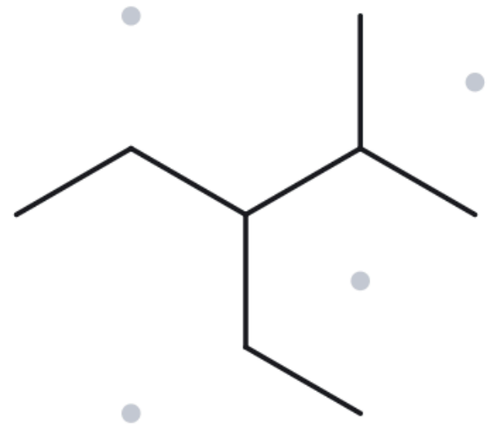
## 1. NAME DECODED

**Parent:**
pentane → a 5-carbon chain

**Substituents:**
methyl on C2; ethyl on C3

**Expected formula:**
5 + 1 + 2 = 8 C; alkane → $C_8H_{18}$

## 2. BOND-LINE STRUCTURE (THE ALKANE)



3-ethyl-2-methylpentane

## 3. FORMULA CHECK

**Counted from the drawing:**
$C_8H_{18}$. The drawing has 8 carbons (5 chain + 1 methyl + 2 ethyl). The chain has 3 + 1 + 1 + 2 + 3 = 10 H, the methyl has 3 H, and the ethyl has 5 H, which gives 18 H

**Alkane:**
$C_nH_{2n+2} = C_8H_{18}$ ✓

## 4. BACK-CHECK

**Returned by The Alkane:**
3-ethyl-2-methylpentane

✓ Same as the given name

- [x] parent length matches
- [x] substituent positions match
- [x] formula agrees
- [x] back-check recorded

# CARD B05 — NAME → STRUCTURE

**Level: DEVELOPING** **GIVEN NAME: 1-chloro-2-methylpropane**

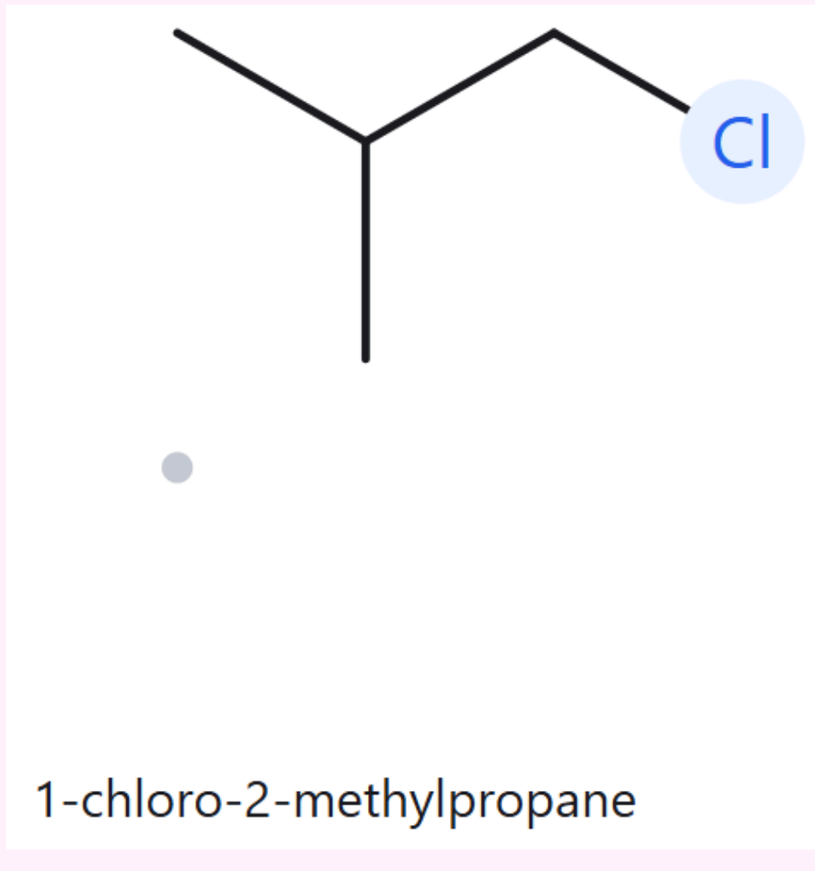
## 1. NAME DECODED

**Parent:**
propane → a 3-carbon chain

**Substituents:**
chloro on C1; methyl on C2

**Expected formula:**
3 + 1 = 4 C; alkane → $C_4H_{10}$; Cl replaces one H → $C_4H_9Cl$

## 2. BOND-LINE STRUCTURE (THE ALKANE)




1-chloro-2-methylpropane

## 3. FORMULA CHECK

**Counted from the drawing:**
$C_4H_9Cl$. The drawing has 4 carbons (3 chain + 1 methyl). The chain has 2 + 1 + 3 = 6 H (C1 carries Cl and 2 H), and the methyl has 3 H, which gives 9 H plus 1 Cl

**Alkane check:**
$C_4H_{10}$ minus one H for Cl = $C_4H_9Cl$ ✓

## 4. BACK-CHECK

**Returned by The Alkane:**
1-chloro-2-methylpropane

✓ Same as the given name

- [x] parent length matches
- [x] substituent positions match
- [x] formula agrees
- [x] back-check recorded

# CARD B06 — NAME → STRUCTURE

**Level: DEVELOPING** **GIVEN NAME: 1,2-dimethylcyclopentane**

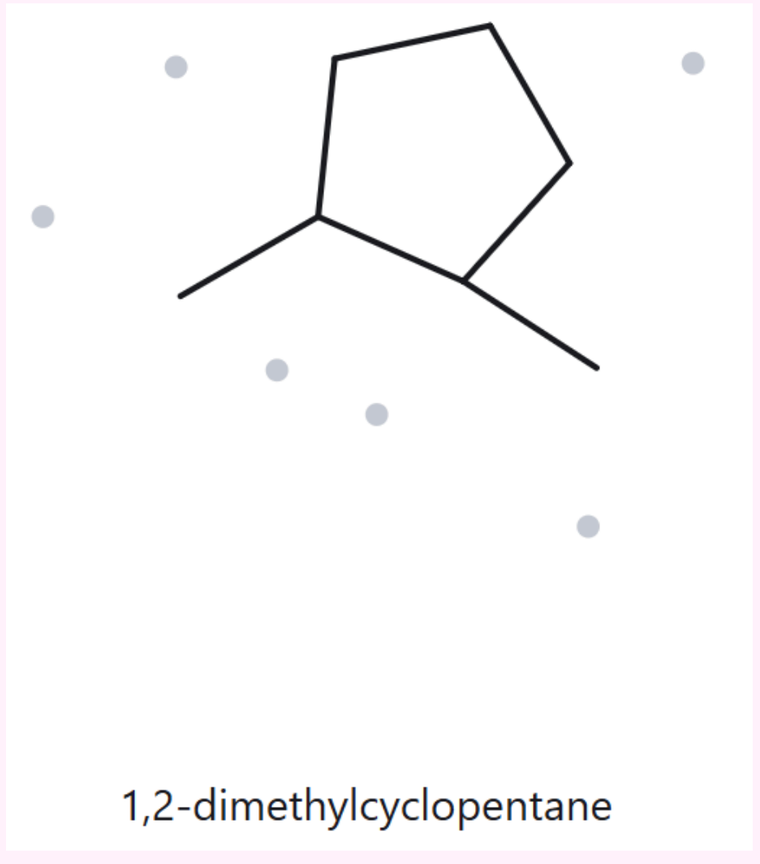
## 1. NAME DECODED

**Parent:**
cyclopentane → a 5-carbon ring

**Substituents:**
methyl on C1; methyl on C2 (adjacent ring carbons)

**Expected formula:**
5 + 2 = 7 C; one ring → $C_7H_{14}$

## 2. BOND-LINE STRUCTURE (THE ALKANE)



1,2-dimethylcyclopentane

## 3. FORMULA CHECK

**Counted from the drawing:**
$C_7H_{14}$. The drawing has 7 carbons (5 ring + 2 methyl). The two substituted ring carbons have 1 H each, the other three ring carbons have 2 H each, and the two methyls have 3 H each: 2 + 6 + 6 = 14 H

**One ring:**
$C_nH_{2n} = C_7H_{14}$ ✓

## 4. BACK-CHECK

**Returned by The Alkane:**
1,2-dimethylcyclopentane

✓ Same as the given name

- [x] parent length matches
- [x] substituent positions match
- [x] formula agrees
- [x] back-check recorded

# CARD B07 — NAME → STRUCTURE

**Level: DEVELOPING** **GIVEN NAME: 4-ethyl-3,3-dimethylheptane**

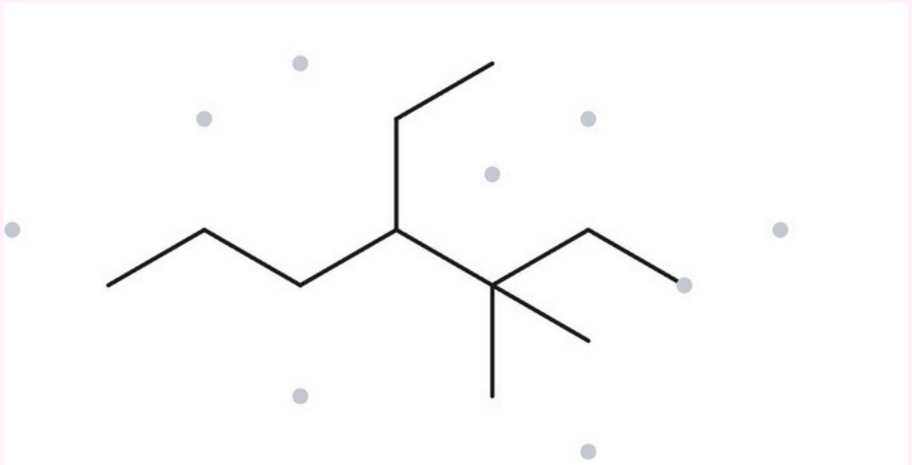
## 1. NAME DECODED

**Parent:**
heptane → a 7-carbon chain

**Substituents:**
two methyl groups, both on C3; ethyl on C4

**Expected formula:**
7 + 2 + 2 = 11 C; alkane → $C_{11}H_{24}$

## 2. BOND-LINE STRUCTURE (THE ALKANE)




## 3. FORMULA CHECK

**Counted from the drawing:**
$C_{11}H_{24}$. The chain has 3 + 2 + 0 + 1 + 2 + 2 + 3 = 13 H (C3 has no H), the two methyls have 6 H, and the ethyl has 5 H, which gives 24 H

**Alkane:**
$C_nH_{2n+2} = C_{11}H_{24}$ ✓

## 4. BACK-CHECK

**Returned by The Alkane:**
4-ethyl-3,3-dimethylheptane

✓ Same as the given name

- [x] parent length matches
- [x] substituent positions match
- [x] formula agrees
- [x] back-check recorded

# CARD B08 — NAME → STRUCTURE

**Level: CHALLENGE** **GIVEN NAME: tert-butylcyclohexane**

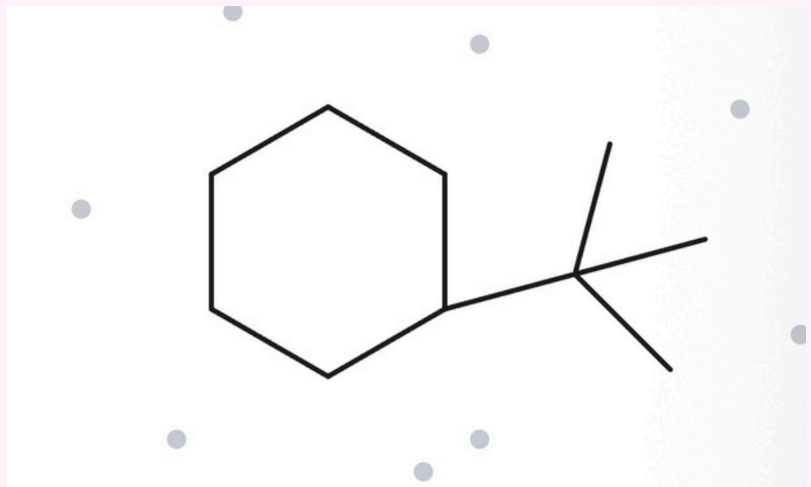
## 1. NAME DECODED

**Parent:**
Cyclohexane → a 6-carbon ring

**Substituents:**
tert-butyl on C1. This is a 4-carbon group, $-C(CH_3)_3$, attached through its central carbon, which bears three methyl groups. A single substituent on a ring needs no locant

**Expected formula:**
6 + 4 = 10 C; ONE RING → $C_{10}H_{20}$

## 2. BOND-LINE STRUCTURE (THE ALKANE)



(1,1-dimethylethyl)cyclohexane

## 3. FORMULA CHECK

**Counted from the drawing:**
$C_{10}H_{20}$. The ring has 5 $CH_2$ (10 H) and 1 CH (1 H) = 11 H. The central carbon of the tert-butyl group has no H, and its three methyls have 9 H, which gives 20 H

**One ring:**
$C_nH_{2n} = C_{10}H_{20}$ ✓

## 4. BACK-CHECK

**Returned by The Alkane:**
(1,1dimethylethyl)cyclohexane

✓ Same compound as the given name, written in systematic substituent form

- [x] parent length matches
- [x] substituent positions match
- [x] formula agrees
- [x] back-check recorded

# CARD B09 — NAME → STRUCTURE

**Level: CHALLENGE** **GIVEN NAME: 3-ethyl-1,1- dimethylcyclohexane**

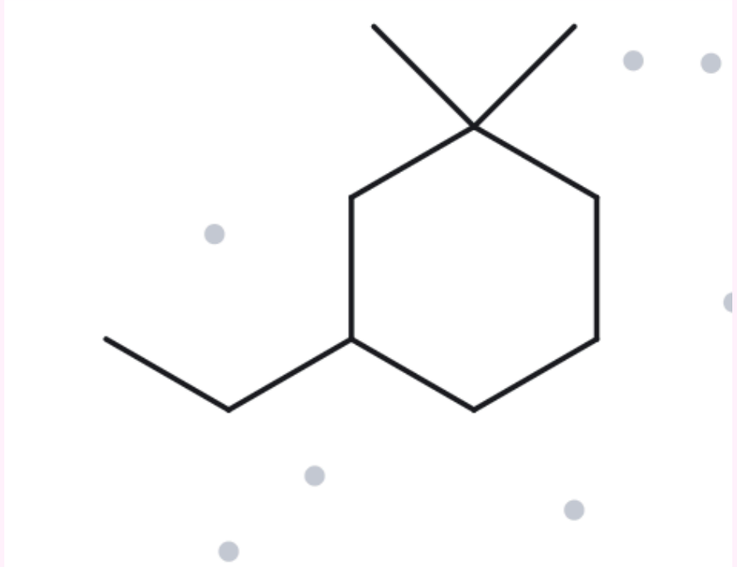
## 1. NAME DECODED

**Parent:**
cyclohexane → a 6-carbon ring

**Substituents:**
two methyl groups, both on C1; ethyl on C3 (one carbon between them, so they are 1,3 to each other)

**Expected formula:**
6 + 2 + 2 = 10 C; one ring → $C_{10}H_{20}$

## 2. BOND-LINE STRUCTURE (THE ALKANE)



3-ethyl-1,1-dimethylcyclohexane

## 3. FORMULA CHECK

**Counted from the drawing:**
$C_{10}H_{20}$. The ring has 0 + 2 + 1 + 2 + 2 + 2 = 9 H (C1 has none, C3 has 1), the two methyls have 6 H, and the ethyl has 5 H, which gives 20 H

**One ring:**
$C_nH_{2n} = C_{10}H_{20}$ ✓

## 4. BACK-CHECK

**Returned by The Alkane:**
3-ethyl-1,1 dimethylcyclohexane

✓ Same as the given name

- [x] parent length matches
- [x] substituent positions match
- [x] formula agrees
- [x] back-check recorded

# CARD B10 — NAME → STRUCTURE

**Level: CHALLENGE** **GIVEN NAME: 2-ethylbutane**
**deliberately incorrect name**

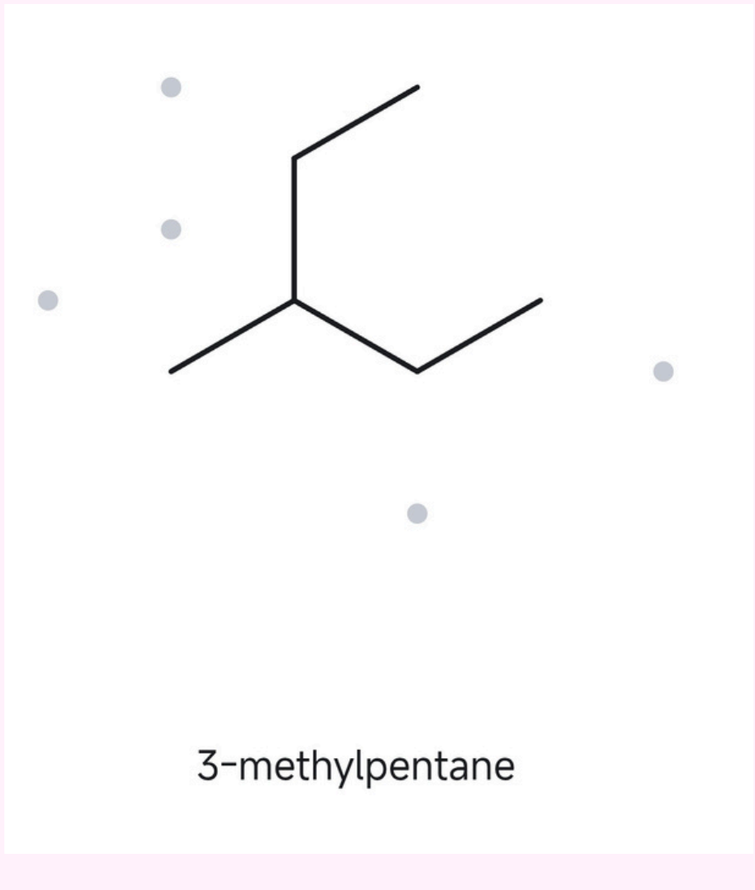
## 1. NAME DECODED

**Parent (as written):**
butane → a 4-carbon chain

**Substituents:**
ethyl on C2

**Expected formula:**
4 + 2 = 6 C; alkane → $C_6H_{14}$.

## 2. BOND-LINE STRUCTURE (THE ALKANE)



3-methylpentane

## 3. FORMULA CHECK

**Counted from the drawing:**
$C_6H_{14}$. The chain has 3 + 1 + 2 + 3 = 9 H (C2 has 1 H), and the ethyl has 5 H, which gives 14 H.

**Alkane:**
$C_nH_{2n+2} = C_6H_{14}$ ✓

## 4. BACK-CHECK

**Returned by The Alkane:**
3-methylpentane

✗ Does not match the given name (the given name breaks a rule)

2-ethylbutane picks a 4-C parent, but a 5-C chain exists through the ethyl branch. The longest chain must be the parent, so it is pentane with a methyl on C3 → 3-methylpentane. Same compound, $C_6H_{14}$.

- [x] parent length matches
- [x] substituent positions match
- [x] formula agrees
- [x] back-check recorded